

Phase 2 – Backend Development & Configurations:

Objective:

The primary objective of **Phase 2** is to engineer the backend relational data architecture for the **Automated Network Request Management System** in ServiceNow.

This phase establishes:

- A dedicated custom table (**Request Management, u_request_management**) to store persistent network connection records, decoupling operational network data from temporary catalog transactions.
- An auto-numbering mechanism (**REQ0001000**) ensuring unique record identification.
- A structured schema comprising 9 custom dictionary fields capturing technical, contact, billing, and fulfillment attributes.
- A balanced 2-column operational form layout.
- A custom platform relationship linking native approvals (**sysapproval_approver**) directly to the custom table with complete audit history tracking.

Data Architecture:

Technical Design:

Standard ServiceNow Service Catalog submissions store transactional information across `sc_request`, `sc_req_item`, and `sc_task`. While effective for generic request ticketing, enterprise network infrastructure requires:

- Long-Term Asset & Connection Registry: A dedicated repository to maintain active, modified, and retired network connections.
- Specialized Governance: Independent Access Control Lists (ACLs) separating end-user access from network engineering operations.
- Seamless Relationship Mapping: Direct relational queries between approvals and customer connection records without cluttering standard platform task tables.

Custom Table Specifications

Table Label: Request Management

Table Name: `u_request_management`

Application Scope: Global

Extends Table: -- None --

Auto-Numbering Settings:

Auto-number: Enabled (true)

Prefix: SKILL

Number: 0001000

Number of Digits: 4

Generated Examples: SKILL1001, SKILL1002, SKILL1003

Data Dictionary (Schema Definition): u_database_tables

Field Label	Column Name	Data Type	Reference Table	Max Length	Mandatory
Database Number	u_database_number	String	-	40	Yes
Requested For	u_Requested_for	String	-	100	Yes
Mobile Number	u_mobile_number	String	-	40	Yes
Total Amount	u_total_amount	String	-	40	Yes
Mode of Payment	u_mode_of_payment	String	-	40	Yes
Type of Connection	u_type_of_connection	String	-	40	Yes
Customer Address	u_customer_address	String	-	255	Yes
Approval Status	u_approval_status	String	-	40	Yes
Assigned To	u_assigned_to	Reference	User [sys_user]	32	No

Form Layout & Usability Design:

LEFT COLUMN (Financial & Identifiers)	RIGHT COLUMN (Customer & Operational Status)
Database Number	Requested For
Mobile Number	Type of Connection
Total Amount	Customer Address
Mode of Payment	Assigned to
	Approval Status

Approval Request Relationship:

System Relationship Architecture:

To integrate approval audit logging with custom records without redundant fields, a ServiceNow **Relationship Record (sys_relationship)** was established.

- **Relationship Name:** Approval Request
- **Applies to Table:** Request Management [u_request_management]
- **Queries from Table:** Approval [sysapproval_approver]
- **Active:** true

Refinement Query Script:

```
(function refineQuery(current, parent) {  
    // Queries approvals where source document points to this specific Database Table record  
    current.addQuery('source_table', parent.getTableName());  
    current.addQuery('document_id', parent.sys_id);  
})(current, parent);
```

Form Related List Integration:

The relationship was added to the form layout via **Configure > Related Lists**:

- **Selected Related List:** Approval Request
- **Exposed Columns:** State, Approver, Comments, Approval for, Created
- **Operational Benefit:** Provides real-time visibility into approver identity, decision timestamps, and audit history directly inside the ticket record.

Step-by-Step Implementation Guide (PDI)

Step 1: Create the Custom Table

1. In the Filter Navigator, navigate to: **System Definition > Tables**.
2. Click **New**.
3. Set **Label**: Request Management, **Name**: u_request_management.
4. In the **Controls** tab:
 - Check **Auto-number** (true).
 - Set **Prefix**: SKILL, **Number**: 1000, **Digits**: 4.
5. Click **Right-click header > Save**.

Step 2: Create the 9 Dictionary Fields

1. On the table record, scroll to the **Columns** related list.
2. Click **New** for each field defined in the Data Dictionary table (Section 3).
3. Click **Update** when all columns are created.

Step 3: Configure Form Layout

1. Open u_request_management.list in the Filter Navigator → click **New**.
2. Right-click the header → **Configure > Form Design**.
3. Set section to **2-Column Layout** and arrange fields according to Section 4.
4. Click **Save**.

Step 4: Create Approval Relationship

1. Navigate to: **System Definition > Relationships** → click **New**.
2. Name: Approval Request.
3. Applies to table: Request Management [u_request_management].
4. Queries from table: Approval [sysapproval_approver].
5. Paste the query script into the **Query with** editor.
6. Click **Submit**.

Step 5: Add Related List to Form

1. Open u_request_management.list → open any record.
2. Right-click header → **Configure > Related Lists**.
3. Move **Approval Request** from *Available* to *Selected*.
4. Click **Save**.

Test Verification & Results:

Test Case	Verification Activity	Expected Outcome	Actual Result	Status
1	Table Accessibility	u_request_management.list opens with column headers	Table renders properly	PASS
2	Auto-Numbering	New record creation generates REQ sequence	Number generated (e.g. SKILL1001)	PASS
3	Field Data Types	Test string, reference, and numeric inputs	Fields validate correct formats	PASS
4	Form Usability	Verify 2-column alignment	Fields positioned cleanly	PASS
5	Approval Tab Rendering	Check form bottom related lists	Approval Request tab renders live	PASS